Data retention and deletion

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask which age clock and which listed content locations the claim covers.

Teach inclusive boundaries, metadata/content separation and preserved failure/retry reports without claiming real secure erasure or legal requirements.

Keep one habit: Model primary/cache actions → inspect explicit backup failures → compare content/audit clocks → assess a fresh temporary-log residual → record scope limits.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of data retention and deletion

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Sandboxing and resource limits. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of sandboxing and resource limits

Reference example, not the only acceptable wording: Teacher explanation: Sandboxing and quotas constrain what a **process** may access or consume. They reduce possible operations but do not verify every result.

The fictional job may read its fixture directory, write its **output** directory, use no network and run for at most 2 seconds with at most 10 queued tasks. Reject excess work before starting it.

Without opening the old lesson, explain one key idea from External side effects and recovery. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of external side effects and recovery

Reference example, not the only acceptable wording: Teacher explanation: External effects can succeed while their acknowledgement is lost. Recovery must inspect committed state before repeating the operation.

The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve compensation as a separate authorised action.

Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of planning and action selection

Reference example, not the only acceptable wording: Teacher explanation: Planning proposes steps; execution must still validate permissions and prerequisites for each step. Replanning cannot invent authority.

For read-only reporting, allowed plan is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review. Reject write or send steps before they run.

Pictured end goal

Explain data retention and deletion using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Case exposure check: this source example appears in ai-3/28, ai-6/20. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Sandboxing and resource limits

External side effects and recovery

Planning and action selection

Use the three-session teaching cycle above, with extra time as needed. Retention and deletion cover all declared stores, including caches and logs. Removing a visible row may leave copies elsewhere. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of data retention and deletion

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach inclusive age rules, copy inventories and policy-correct evidence retention

> Separate main backup uncertainty, pictured explicit backup policy and the current independent 4-day task.

Definitions and complete fictional retention policy

Retention and deletion cover all declared stores, including caches and logs. Removing a visible row may leave copies elsewhere.

Toy policy retains session content for 7 days and deletes it when age is at least 7. Audit metadata may remain for 30 days without content. Track primary storage, cache and backup policy separately.

Main fictional retention policy	Exact classroom rule
Age measurement	Supplied nonnegative whole-day ages at one fixed snapshot
Session-content boundary	Age at least 7 days: removal due; age below 7: retain for now
Main listed content stores	Primary and cache, both initially holding the three supplied records
Backups in main case	Need their separately stated expiry process; not inspected by this fixture
Audit metadata	May remain under a separate 30 day policy, without transcript content
Audit age clock	Age since the audit event was created, not the original transcript age
Verification	Check due content absent from every listed store in the modeled scope
Code scope	Transform copies of fictional dictionaries; no actual storage deletion or secure erasure
Policy status	These toy lifetimes are supplied exercise rules, not legal retention requirements

Main copy ledger and checked reasoning

Main record ID	Content age	Primary/cache copies initially	Required modeled action
S6	6	One in each of2 stores	Retain both content copies for now
S7	7	One in each of2 stores	Remove both: equality reaches deletion boundary
S9	9	One in each of2 stores	Remove both: above boundary
After successful scoped simulation	S6 remains in both stores	Four due entries removed	Only primary/cache postconditions checked
Main backups	No snapshot supplied	Separate expiry policy still needed	Do not claim global deletion
New allowed audit events	Event age 0	No transcript payload	Audit age 29: retain; age 30: due

Require inclusive deletion at age 7, not only ages above 7. Main primary/cache have S6/S7/S9, so four due S7/S9 copies are removed in the transformed model; S6 stays, and backup expiry remains unverified. Audit events contain IDs/action/location/event age only, with a separate 30 day clock.

Picture T2 needs both primary and its listed backup removed; one failed backup leaves scoped_complete false, and the prior failure report survives a retry. The Python simulation changes copied dictionaries and cannot prove actual erasure or complete store discovery. The selected current independent case uses a 4-day boundary and a temporary-log residual; earlier 5-day wording is only retained practice.

Data retention and deletion — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Record ages are 6, 7, 9 days. Cache contains copies of all three.	Toy policy retains session content for 7 days and deletes it when age is at least 7.	Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry process.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry **process**.

Qualification: This fictional retention exercise is not a statement of legal retention requirements.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Record ages are 6, 7, 9 days. Cache contains copies of all three.

Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry **process**.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only policy-permitted audit metadata without expired content so a later revision can be compared honestly.

Unknown-age, audit-clock and residual-copy probes

Retention or inventory probe	Expected interpretation
Content age 6 under7 day rule	Retain for now
Content age 7or9	Removal due from every applicable listed content store
Primary removed, listed backup remains	Scoped deletion incomplete; do not report success
Unknown record age in a listed store	Reject incomplete manifest rather than guess retention status
No listed-store inventory	Reject empty/incomplete fixture; do not claim vacuous global deletion
New content-free audit event	Own age clock starts at creation; its 30 day lifetime is separate
Expired content copied into audit payload	Violates the stated no-content metadata policy
Same modeled removal retried	Check remaining copies and retain prior failures; do not overwrite history
All listed copies absent in the model	Scoped postcondition only, not proof of unknown-store deletion or physical erasure

Separate picture inventory and failure/retry scope

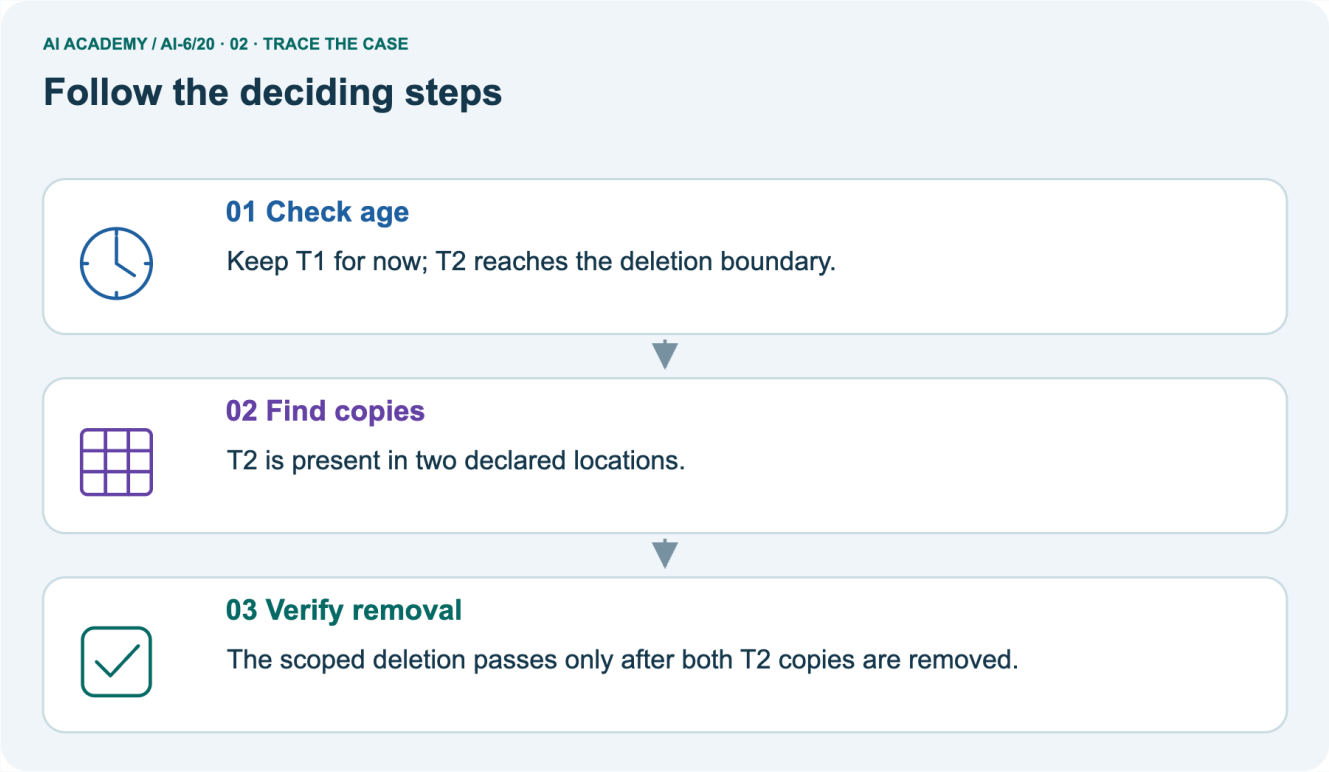
Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups.

T1 is six days old. T2 is seven days old with one backup. Scope lists primary storage and that backup only.

The main example checks primary/cache and leaves backup expiry unverified. The picture has a separate explicit scope: primary storage and the one listed backup of T2, both covered by its fictional 7-day content policy. T1 at age 6 is retained; T2 at age 7 requires removal from two locations. If primary removal succeeds but the backup remains, record incomplete deletion and that residual location.

A separate retry can clear the remaining modeled copy while the failed report stays in the history. This establishes no removal from unknown locations or real physical media.

Picture trace and incomplete-backup key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check age	Keep T1 for now; T2 reaches the deletion boundary.
2. Find copies	T2 is present in two declared locations.
3. Verify removal	The scoped deletion passes only after both T2 copies are removed.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Retention requires an age rule and an inventory. Deleting the visible copy alone does not demonstrate removal of listed backups.

The inventory cannot prove deletion from unknown locations; document scope and investigate untracked copies.

Change: Primary deletion succeeds but backup deletion fails.

Prediction: Record incomplete deletion and follow the retry or escalation procedure.

Reason: One required copy remains in the declared scope.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Retention period	The declared lifetime of stored information
Deletion boundary	The age at which removal applies
Residual copy	A remaining duplicate in another store

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

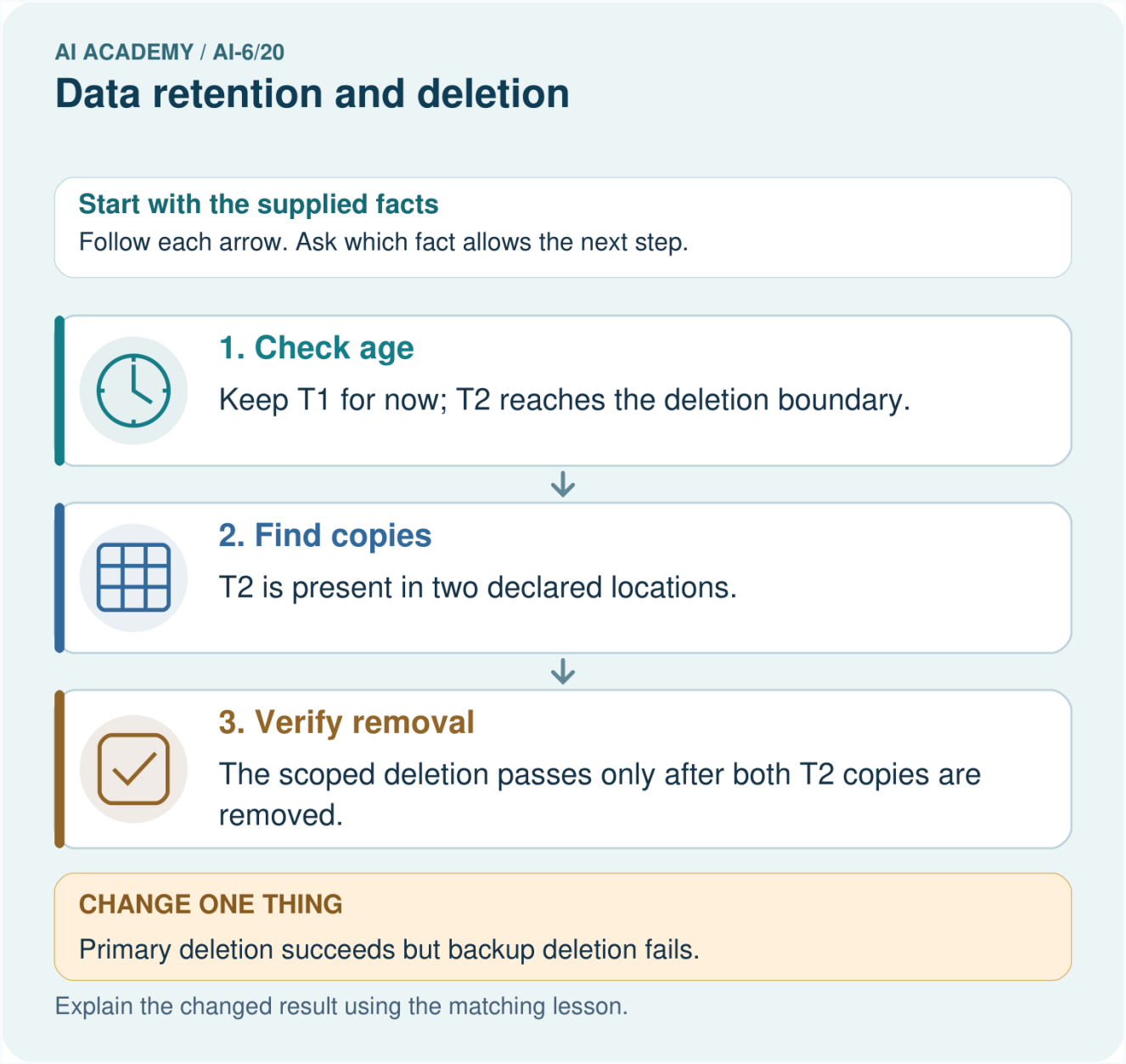
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Data retention and deletion

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Record ages are 6, 7, 9 days. Cache contains copies of all three. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry **process**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Data retention and deletion. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: This fictional retention exercise is not a statement of legal retention requirements.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Apply the seven-day boundary to T1 and T2. How many T2 locations need verification, and may deletion be called complete if its backup remains?

Keep T1 for now; T2 reaches the **deletion boundary**. T2 is present in two declared locations. The scoped deletion passes only after both T2 copies are removed.

Changed case: Record incomplete deletion and follow the retry or escalation procedure. One required copy remains in the declared scope.

Changed-case reasoning: Record incomplete deletion and follow the retry or escalation procedure. One required copy remains in the declared scope.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main age 6 content is retained; ages 7 and 9 are due in primary/cache, while backups need their separate expiry policy. Picture T2 has two listed locations and remains incomplete if its backup survives. Audit metadata uses its own 30-day clock and contains no transcript payload. Selected current 4-day rule: retain X at age 3; Y at age 4 and Z at age 5 are due.

A full-content Y copy in a temporary log still violates the policy after its primary copy disappears. Use the current 4-day task rather than earlier 5-day practice, and keep only permitted metadata as failure evidence.

Diagnose the current misconceptions

The worked result establishes every future case: This fictional retention exercise is not a statement of legal retention requirements.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Fictional retention policy deletes records at age >=4 days. X is3 days, Y4, Z5. A copy of Y remains in a temporary log after its main record is deleted. Evaluate compliance.

Teacher explanation: Retain X at age 3 days. Delete Y at age 4 and Z at age 5 from every store covered by the fictional policy. The temporary log still contains a copy of Y, so deleting its main record alone is incomplete. Remove that content through the declared deletion process and verify the copies; do not treat this fictional rule as a legal retention requirement.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Retention and deletion cover all declared stores, including caches and logs. Removing a visible row may leave copies elsewhere.

Toy policy retains session content for 7 days and deletes it when age is at least 7. Audit metadata may remain for 30 days without content. Track primary storage, cache and backup policy separately.

W2 Trace the worked case

Record ages are 6, 7, 9 days. Cache contains copies of all three. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry **process**.

W3 Audit the limit

A report omits this qualification: This fictional retention exercise is not a statement of legal retention requirements. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: This fictional retention exercise is not a statement of legal retention requirements.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. This fictional retention exercise is not a statement of legal retention requirements.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry **process**.

Qualification: This fictional retention exercise is not a statement of legal retention requirements.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Retention is5 days, inclusive **deletion boundary**. Ages 4, 5, 6; a log stores complete content. Trace required action.

Teacher explanation: Age 4 retained; age 5/6 content removed from declared stores. Full-content logs violate minimisation and need policy-compliant handling.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Age 4 retained; age 5/6 content removed from declared stores. Full-content logs violate minimisation and need policy-compliant handling. Require intermediate reasoning and the stated scope.

Fresh retry: Retention 3 days; records ages 2, 3. Expected reasoning: Age 2 retained; age 3 **deletion boundary** applies. Document residual copies under the backup policy.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled data retention and deletion **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of data retention and deletion.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Caching and stale information. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/21, ai-6/23, ai-6/26. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of data retention and deletion

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Record ages are 6, 7, 9 days. Cache contains copies of all three. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Retain age 6; delete content ages 7 and 9 from primary and cache under the policy. Retain only allowed audit metadata; backups need their stated expiry **process**.

Check every learner before discussion: Use the worked case for Data retention and deletion. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: This fictional retention exercise is not a statement of legal retention requirements.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Primary deletion succeeds but backup deletion fails. Require a **prediction** and a reason before comparing. Expected result: Record incomplete deletion and follow the retry or escalation procedure.

Reason: One required copy remains in the declared scope.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Data retention and deletion: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

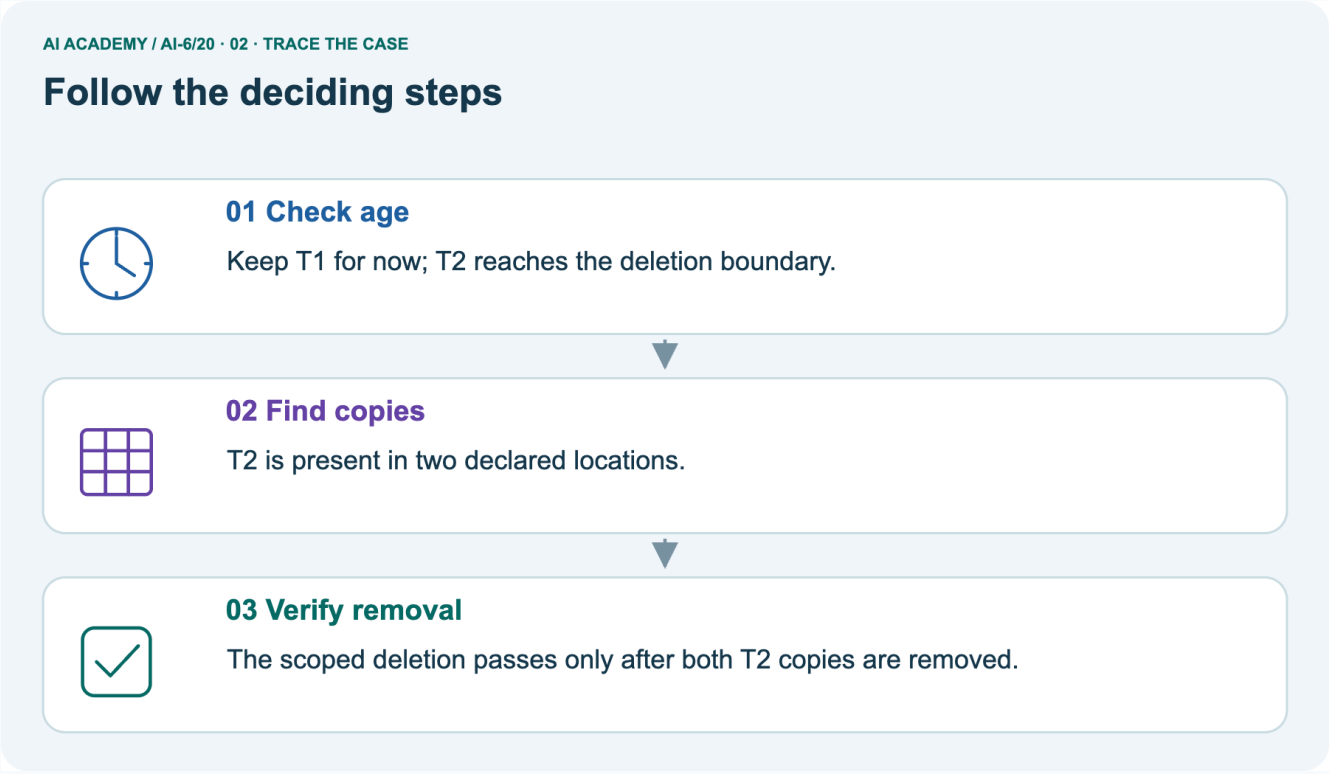
Delete every listed copy. A fictional policy deletes transcripts at age seven days or older, including listed backups. Supplied facts: T1 is six days old. T2 is seven days old with one backup.

Scope lists primary storage and that backup only.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



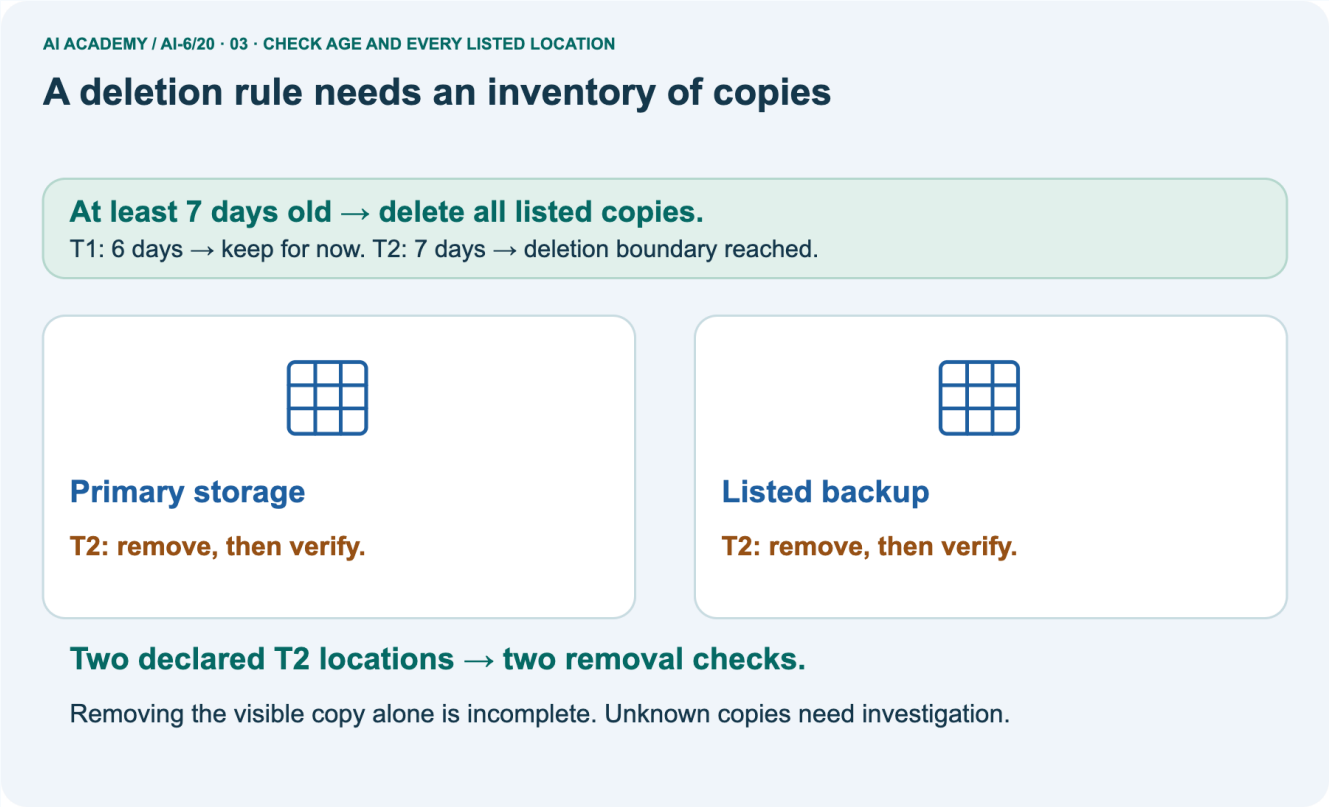
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Check age: Keep T1 for now; T2 reaches the **deletion boundary**. → Find copies: T2 is present in two declared locations. → Verify removal: The scoped deletion passes only after both T2 copies are removed.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. A deletion rule needs an inventory of copies



Follow the labelled picture with the matching explanation. Open the vector diagram

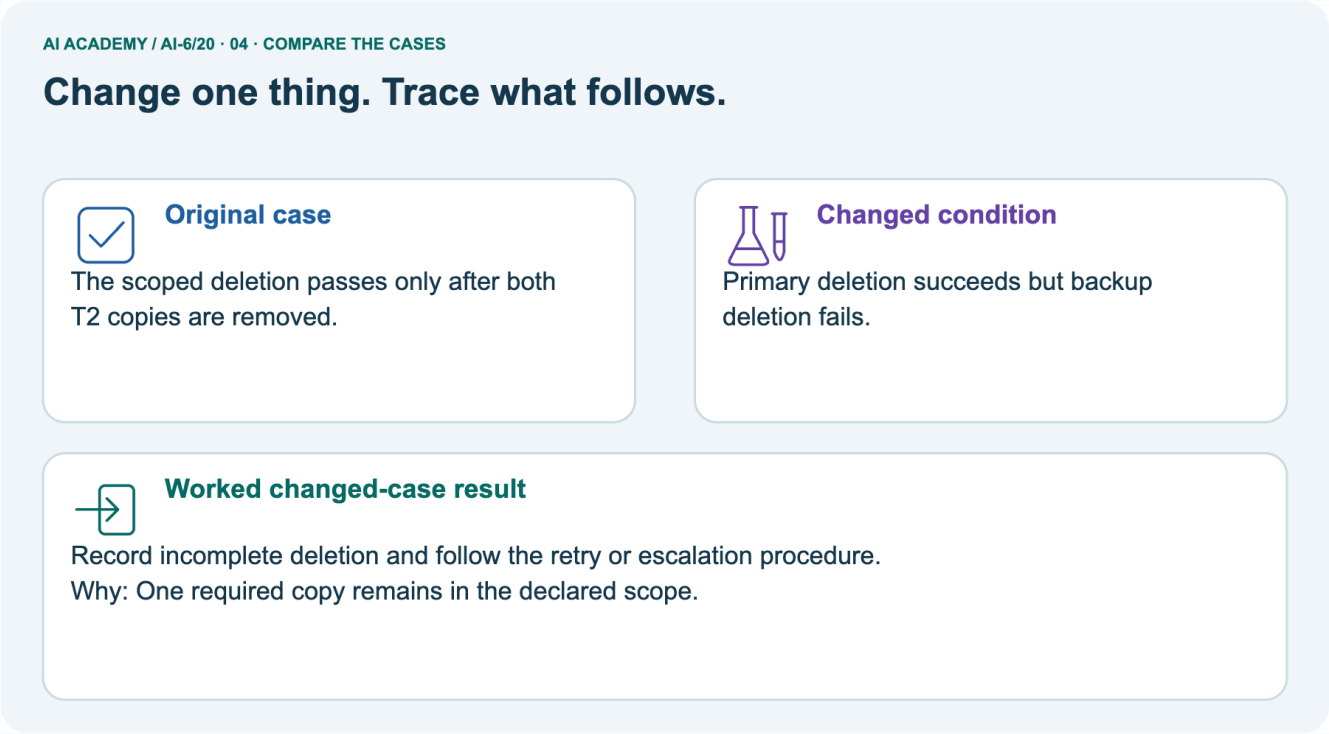
The toy retention boundary is seven days or older. Keep six-day T1 for now. Seven-day T2 appears in primary storage and one listed backup; both locations need removal verification.

This diagram specifies required checks, not an actual deletion run.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The scoped deletion passes only after both T2 copies are removed. Changed condition: Primary deletion succeeds but backup deletion fails.

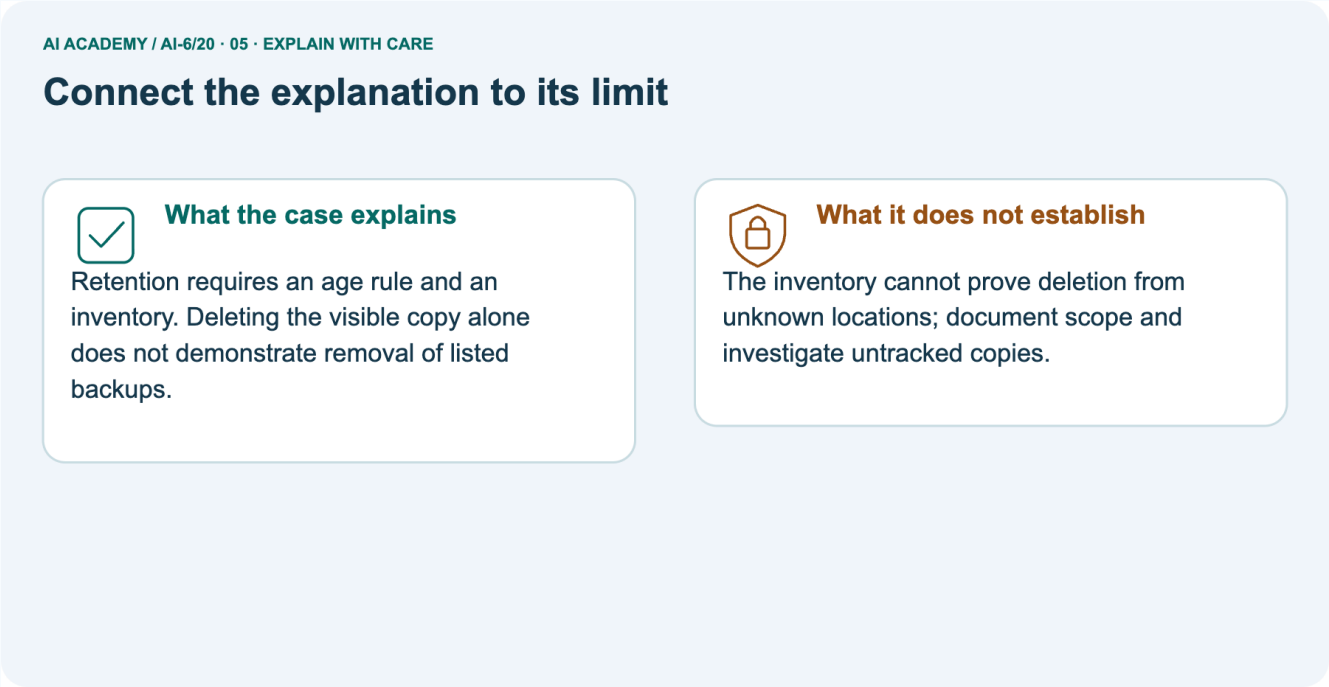
Expected result: Record incomplete deletion and follow the retry or escalation procedure. Reason: One required copy remains in the declared scope.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: One required copy remains in the declared scope.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Retention requires an age rule and an inventory. Deleting the visible copy alone does not demonstrate removal of listed backups.

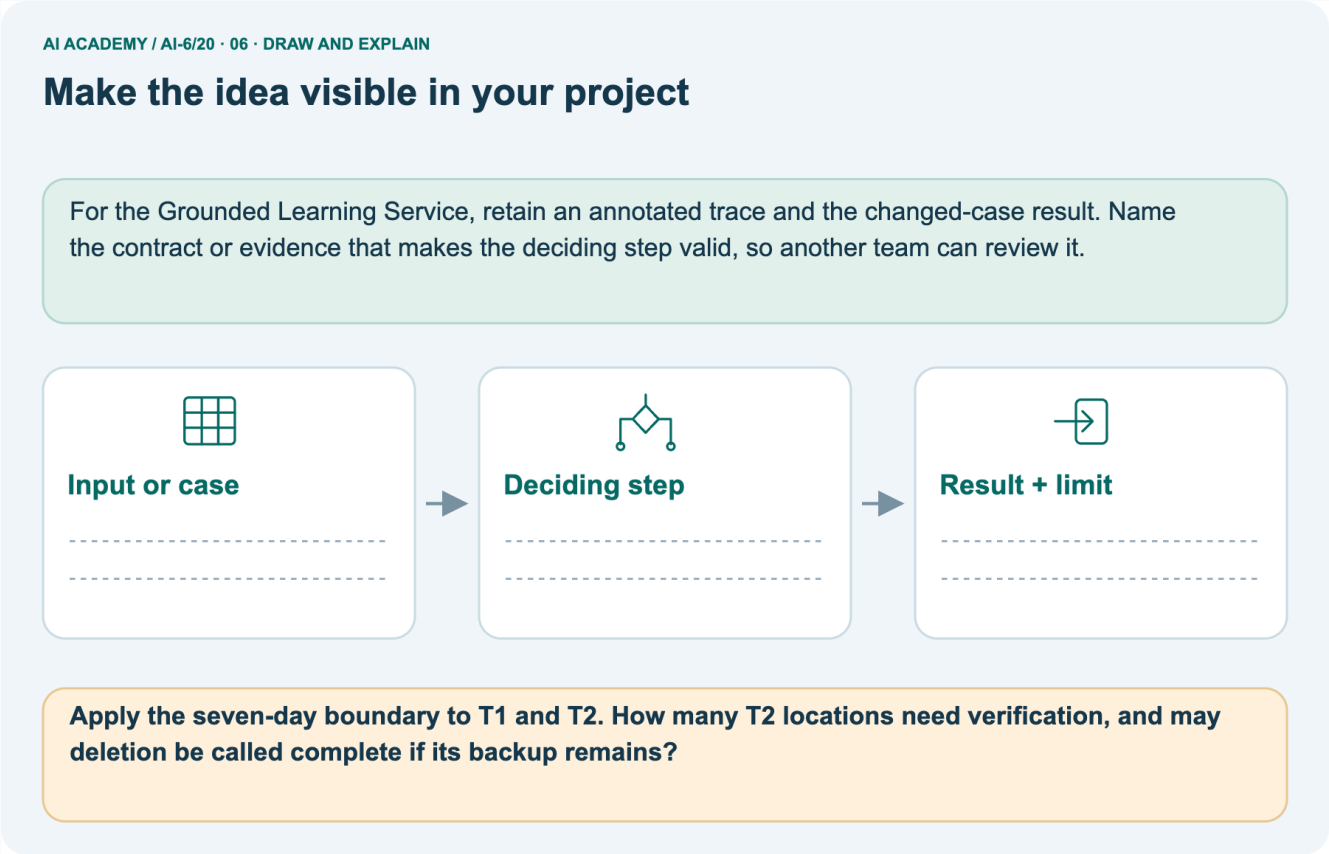
Limit: The inventory cannot prove deletion from unknown locations; document scope and investigate untracked copies.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: The inventory cannot prove deletion from unknown locations; document scope and investigate untracked copies.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the seven-day boundary to T1 and T2. How many T2 locations need verification, and may deletion be called complete if its backup remains?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Keep T1 for now; T2 reaches the **deletion boundary**. T2 is present in two declared locations. The scoped deletion passes only after both T2 copies are removed.

Changed case: Record incomplete deletion and follow the retry or escalation procedure. One required copy remains in the declared scope.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

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▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.